GEOMETRIC REALIZATIONS OF ABSOLUTE GALOIS GROUPS IN PERFECTOID GEOMETRY

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Abstract. We generalize Weinstein's geometric realization of absolute Galois groups to a class of complete rank-one nonarchimedean fields of mixed characteristic. Let E_∞/E be a Galois algebraic extension whose completion $L = \widehat{E_\infty}$ is perfectoid, and let $C = \widehat{\overline{E}}$. Using the product of $\mathrm{Spa}(C^\flat)$ and $\mathrm{Spa}(L^\flat)$ in the category of perfectoid spaces, a partial-Frobenius quotient, and Galois descent, we construct a sheaf Z_E on the pro-étale site over C such that

$$\mathbf{F\acute{E}t}(Z_E) \simeq \mathbf{F\acute{E}t}(E).$$

Consequently,

$$\pi_1^{\text{ét}}(Z_E) \cong G_E$$

after choosing a geometric base point. We apply the construction to the two-dimensional local field $\mathbb{Q}_p\{\{X\}\}$ via its natural Kummer tower.

1. Introduction

Let $E/\mathbb{Q}_p$ be a finite extension. In [6], Weinstein constructed a geometric object Z_E, defined over a complete algebraically closed extension C/E, whose étale fundamental group recovers the absolute Galois group of E. More precisely, he proved an equivalence of Galois categories

$$\mathbf{F\acute{E}t}(Z_E) \simeq \mathbf{F\acute{E}t}(E).$$

The point of this construction is not merely to realize G_E as a fundamental group: the Fargues–Fontaine curve already provides such a realization. Rather, Weinstein's construction produces an object defined over an algebraically closed nonarchimedean field, with the arithmetic information encoded in its geometry and group action.

The purpose of this paper is to extend this construction beyond ordinary local fields. Our main example is the two-dimensional local field

$$E = \mathbb{Q}_p\{\{X\}\}.$$

More generally, the same construction applies to a complete rank-one nonarchimedean field of mixed characteristic admitting a Galois algebraic extension whose completion is perfectoid; see Section 6 for the precise hypotheses.

Theorem A. *Let*

$$E = \mathbb{Q}_p\{\{X\}\}.$$

There exists a geometric object Z_E, defined over the complete algebraically closed field $\widehat{\overline{E}}$, such that

$$\mathbf{F\acute{E}t}(Z_E) \simeq \mathbf{F\acute{E}t}(E).$$

In particular, after choosing a geometric base point,

$$\pi_1^{\text{ét}}(Z_E) \cong G_E.$$

More generally, the same conclusion holds for every complete rank-one nonarchimedean field E of mixed characteristic satisfying the hypotheses of Theorem 6.1.

The organization of the paper is the following: In Section 2, we recall the construction of the adic Fargues–Fontaine curve and the basic results which will be used later. Section 3 reviews Weinstein's realization of G_E for an ordinary local field. In Section 4, we review Kedlaya's base change lemma, which forms the main geometric input of our construction, following from the universal geometric simple connectivity theorem. Section 5 contains the two arithmetic descent statements needed later: invariance of finite étale covers under completion and infinite Galois descent. Finally, in Section 6 we construct the space Z_E, prove the main theorem by combining these ingredients, and apply the general construction to $E = \mathbb{Q}_p\{\{X\}\}$.

2. Preliminaries on the Fargues–Fontaine curve

Let E be a local field of residue characteristic p, with ring of integers $\mathcal{O}_E$, residue field $k_E \cong \mathbb{F}_q$, and a fixed uniformizer π. Let F be an algebraically closed nonarchimedean field of characteristic p equipped with an embedding $k_E \hookrightarrow F$. We fix a pseudo-uniformizer $\varpi_F \in \mathcal{O}_F$. For $a \in \mathbb{Q}_{>0}$, choose $\varpi_F^a \in F$ such that $|\varpi_F^a| = |\varpi_F|^a$.

We briefly recall the construction of the adic Fargues–Fontaine curve associated to the pair (F, E).

2.1. **The mixed-characteristic case.** Suppose first that $\mathrm{char}(E) = 0$. Put

$$A_{\mathrm{inf},F,E} := W_{\mathcal{O}_E}(\mathcal{O}_F) := W(\mathcal{O}_F) \otimes_{W(k_E)} \mathcal{O}_E.$$

Since F is perfect, every element of $A_{\mathrm{inf},F,E}$ admits a unique Teichmüller expansion

$$x = \sum_{n\geq 0} \pi^n [x_n], \qquad x_n \in \mathcal{O}_F.$$

We equip $A_{\mathrm{inf},F,E}$ with the $(\pi, [\varpi_F])$-adic topology and define

$$\mathcal{Y}_{F,E} := \mathrm{Spa}(A_{\mathrm{inf},F,E}, A_{\mathrm{inf},F,E}) \setminus V\big(\pi[\varpi_F]\big).$$

For a closed interval $I = [a, b] \subset (0, \infty)$ with $a, b \in \mathbb{Q}$, define

$$\mathcal{Y}_{F,E}^I := U\left(\frac{\{\pi, [\varpi_F^a]\}}{[\varpi_F^a]}, \frac{\{\pi, [\varpi_F^b]\}}{\pi}\right) = \left\{x \in \mathcal{Y}_{F,E} \,\middle|\, |[\varpi_F^b](x)| \leq |\pi(x)| \leq |[\varpi_F^a](x)|\right\}.$$

As I varies, these rational subsets cover $\mathcal{Y}_{F,E}$.

Let $R_{F,E}^I$ be the π-adic completion of

$$A_{\mathrm{inf},F,E}\left[\frac{\pi}{[\varpi_F^a]}, \frac{[\varpi_F^b]}{\pi}\right],$$

and put

$$B_{F,E}^I := R_{F,E}^I[1/\pi],$$

with $B_{F,E}^{I,+}$ the integral closure of $R_{F,E}^I$ in $B_{F,E}^I$. Then

$$\mathcal{Y}_{F,E}^I = \mathrm{Spa}\left(B_{F,E}^I, B_{F,E}^{I,+}\right).$$

The q-power Frobenius of F induces an automorphism

$$\varphi : A_{\mathrm{inf},F,E} \longrightarrow A_{\mathrm{inf},F,E}, \qquad \sum_{n\geq 0} \pi^n [x_n] \longmapsto \sum_{n\geq 0} \pi^n [x_n^q],$$

and hence an automorphism of $\mathcal{Y}_{F,E}$.

2.2. **The equal-characteristic case.** Suppose now that $\mathrm{char}(E) = p$. After choosing π, we identify

$$E \simeq k_E((\pi)), \qquad \mathcal{O}_E \simeq k_E[[\pi]].$$

In this case put

$$A_{\mathrm{inf},F,E} := \mathcal{O}_F[[\pi]],$$

equipped with the (ϖ_F, π)-adic topology, and define

$$\mathcal{Y}_{F,E} := \mathrm{Spa}\big(\mathcal{O}_F[[\pi]], \mathcal{O}_F[[\pi]]\big) \setminus V(\pi\varpi_F).$$

For $I = [a, b] \subset (0, \infty)$ with $a, b \in \mathbb{Q}$, define

$$\mathcal{Y}^I_{F,E} := U\left(\frac{\{\pi, \varpi_F^a\}}{\varpi_F^a}, \frac{\{\pi, \varpi_F^b\}}{\pi}\right) = \left\{x \in \mathcal{Y}_{F,E} \,\middle|\, |\varpi_F^b(x)| \le |\pi(x)| \le |\varpi_F^a(x)|\right\}.$$

Again these rational subsets cover $\mathcal{Y}_{F,E}$. Let $R^I_{F,E}$ be the π-adic completion of

$$\mathcal{O}_F[[\pi]]\left[\frac{\pi}{\varpi_F^a}, \frac{\varpi_F^b}{\pi}\right],$$

put $B^I_{F,E} := R^I_{F,E}[1/\pi]$, and let $B^{I,+}_{F,E}$ be the integral closure of $R^I_{F,E}$ in $B^I_{F,E}$. Then

$$\mathcal{Y}^I_{F,E} = \mathrm{Spa}\left(B^I_{F,E}, B^{I,+}_{F,E}\right).$$

The q-power Frobenius on the coefficients induces

$$\varphi : \mathcal{O}_F[[\pi]] \longrightarrow \mathcal{O}_F[[\pi]], \qquad \sum_{n\ge 0} x_n\pi^n \longmapsto \sum_{n\ge 0} x_n^q\pi^n,$$

and hence an automorphism of $\mathcal{Y}_{F,E}$.

2.3. **The annular covering.** The rings $B^I_{F,E}$ admit an equivalent description as extended Robba rings. Put

$$B^b_{F,E} = \begin{cases} A_{\mathrm{inf},F,E}\left[\dfrac{1}{\pi}, \dfrac{1}{[\varpi_F]}\right], & \mathrm{char}(E) = 0, \\ A_{\mathrm{inf},F,E}\left[\dfrac{1}{\pi}, \dfrac{1}{\varpi_F}\right], & \mathrm{char}(E) = p. \end{cases}$$

Fix a normalization $0 < |\pi|_E < 1$. For $t > 0$, define

$$\lambda_t\left(\sum_{n\gg -\infty} \pi^n[x_n]\right) = \sup_n |\pi|_E^n |x_n|^t$$

in mixed characteristic, and

$$\lambda_t\left(\sum_{n\gg -\infty} x_n\pi^n\right) = \sup_n |\pi|_E^n |x_n|^t$$

in equal characteristic. For $I = [a, b]$, the ring $B^I_{F,E}$ is the completion with respect to the norms λ_t for $t \in I$. Since $t \mapsto \log \lambda_t(x)$ is convex, it is equivalently the completion for the Banach norm

$$\lambda_I(x) := \max\{\lambda_a(x), \lambda_b(x)\}.$$

The rings $B_{F,E}^I$ are strongly noetherian. In particular,

$$\mathcal{Y}_{F,E} = \varinjlim_I \mathcal{Y}_{F,E}^I = \varinjlim_I \operatorname{Spa}\left(B_{F,E}^I, B_{F,E}^{I,+}\right)$$

is an adic space. The Frobenius satisfies

$$\varphi : B_{F,E}^I \longrightarrow B_{F,E}^{I/q}$$

and induces an automorphism of $\mathcal{Y}_{F,E}$.

The action of $\varphi^{\mathbb{Z}}$ on $\mathcal{Y}_{F,E}$ is properly discontinuous. We define the adic Fargues–Fontaine curve by

$$\mathcal{X}_{F,E} := \mathcal{Y}_{F,E}/\varphi^{\mathbb{Z}}.$$

The quotient $\mathcal{X}_{F,E}$ is a well-defined adic space.

2.4. **Classical results on the Fargues–Fontaine curve.** We recall some classical results on the Fargues–Fontaine curve. For $\lambda \in \mathbb{Q}$, let $\mathcal{O}(\lambda)$ denote the stable vector bundle of slope λ on $\mathcal{X}_{F,E}$. If $\lambda = d/h$ with $(d,h) = 1$ and $h > 0$, then $\mathcal{O}(\lambda)$ has rank h and degree d.

The classification of vector bundles in mixed characteristic is due to Fargues [1, Théorème 3.1]; the equal-characteristic analogue is due to Hartl–Pink [3, Theorem 11.1]. We use the uniform formulation of Fargues–Fontaine [2, Théorème 8.2.10]; see also [5, Theorem 4.6].

Theorem 2.1. *Every vector bundle $\mathcal{E}$ on $\mathcal{X}_{F,E}$ admits a decomposition*

$$\mathcal{E} \cong \bigoplus_{i=1}^n \mathcal{O}(\lambda_i), \qquad \lambda_i \in \mathbb{Q},$$

where the multiset of slopes $\{\lambda_1, \dots, \lambda_n\}$ is uniquely determined by $\mathcal{E}$.

We will also use the following computation of global sections. In mixed characteristic this is [6, Proposition 3.3.3]; the same statement in both characteristics is contained in [5, Lemma 4.5].

Proposition 2.2. *For $\lambda \in \mathbb{Q}$,*

$$H^0\big(\mathcal{X}_{F,E}, \mathcal{O}(\lambda)\big) \neq 0 \quad \Longleftrightarrow \quad \lambda \geq 0.$$

Moreover,

$$H^0\big(\mathcal{X}_{F,E}, \mathcal{O}_{\mathcal{X}_{F,E}}\big) = E.$$

Combining these results with the trace pairing gives the geometric simple connectivity of the Fargues–Fontaine curve. In mixed characteristic this is [6, Theorem 3.4.3]; the same argument applies in equal characteristic using the classification above.

Theorem 2.3. *Base extension induces an equivalence*

$$\mathbf{F\acute{E}t}(E) \xrightarrow{\sim} \mathbf{F\acute{E}t}\big(\mathcal{X}_{F,E}\big).$$

In particular, after choosing a geometric point,

$$\pi_1^{\text{ét}}\big(\mathcal{X}_{F,E}\big) \cong G_E.$$

In addition, we record the following identification, which will be used in Section 4.

Proposition 2.4. *Suppose that* $\mathrm{char}(E) = p$*, so that* $E = \mathbb{F}_q((\pi))$*, and let* A *be a Banach* E*-algebra. Let* R^+ *be the completion of*

$$\mathcal{O}_F \otimes_{\mathbb{F}_q} A^\circ$$

for the (ϖ_F, π)*-adic topology, and put*

$$R = R^+[\varpi_F^{-1}, \pi^{-1}].$$

Then there is a canonical identification

$$\mathcal{Y}_{F,E} \times_E \mathrm{Spa}(A, A^\circ) \cong \left\{ v \in \mathrm{Spa}(R, R^+) \,\middle|\, |\varpi_F(v)| < 1,\ |\pi(v)| < 1 \right\}.$$

This follows from the identification

$$\mathcal{Y}_{F,E} \cong \{x \in \mathrm{Spa}(F, \mathcal{O}_F) \times_{\mathbb{F}_q} \mathrm{Spa}(E, \mathcal{O}_E) \mid |\varpi_F(x)|,\ |\pi(x)| < 1\}.$$

3. Realizing G_E as a fundamental group

We review the construction of Weinstein [6, §3.5]. Throughout this section, let $E/\mathbb{Q}_p$ be a finite extension with residue field $k_E = \mathbb{F}_q$ and uniformizer π, and let C be a complete algebraically closed extension of E.

3.1. The Lubin–Tate universal cover.

Choose a power series

$$f(T) \in T\mathcal{O}_E[[T]], \qquad f(T) \equiv T^q \pmod{\pi},$$

and let $H_E = \mathrm{Spf}\, \mathcal{O}_E[[T]]$ be the Lubin–Tate formal $\mathcal{O}_E$-module associated to f, so that

$$[\pi]_{H_E}(T) = f(T).$$

Choose a compatible sequence of Lubin–Tate torsion points $t_1, t_2, \ldots$ satisfying

$$f(t_1) = 0, \qquad f(t_{n+1}) = t_n,$$

and put

$$E_n := E(t_n), \qquad E_\infty := \bigcup_{n \geq 1} E_n.$$

The completion $\widehat{E}_\infty$ is a perfectoid field.

Let

$$\widetilde{H}_E := \varprojlim_{[\pi]} H_E$$

be the universal cover of H_E. This is a formal E-vector space. Put

$$H_{E,0} := H_E \otimes_{\mathcal{O}_E} \mathbb{F}_q, \qquad \widetilde{H}_{E,0} := \widetilde{H}_E \otimes_{\mathcal{O}_E} \mathbb{F}_q.$$

Since $[\pi]_{H_{E,0}}(T) = T^q$, one has

$$\widetilde{H}_{E,0} \cong \mathrm{Spf}\, \mathbb{F}_q[[T^{1/q^\infty}]].$$

Moreover, Weinstein constructs an $E^\times$-equivariant isomorphism

$$\widetilde{H}_{E,0} \cong \mathrm{Spf}\, \mathcal{O}_{\widehat{E}_\infty^\flat},$$

where $\mathcal{O}_E^\times$ acts on $\widehat{E}_\infty$ through local class field theory and π acts on $\widehat{E}_\infty^\flat$ by the q-power Frobenius [6, Proposition 3.5.3].

Let

$$\widetilde{H}_{E,C}^{\mathrm{ad}} := \left(\widetilde{H}_E \widehat{\otimes}_{\mathcal{O}_E} \mathcal{O}_C \right)_\eta^{\mathrm{ad}}, \qquad \widetilde{H}_{E,C}^{\mathrm{ad},*} := \widetilde{H}_{E,C}^{\mathrm{ad}} \setminus \{0\}.$$

Let $\varpi_C^\flat \in \mathcal{O}_{C^\flat}$ be a pseudo-uniformizer, and let $t \in \mathcal{O}_{\widehat{E}_\infty^\flat}$ be the pseudo-uniformizer arising from the compatible Lubin–Tate tower. By [6, Proposition 3.5.4], there are $E^\times$-equivariant isomorphisms

$$\left(\widetilde{H}_{E,C}^{\mathrm{ad},*}\right)^\flat \cong \left(\mathcal{Y}_{C^\flat,E}\widehat{\otimes}_E\widehat{E}_\infty\right)^\flat \cong \mathrm{Spa}(R,R^+),$$

where

$$R^+ := \mathcal{O}_{C^\flat}\widehat{\otimes}_{\mathbb{F}_q}\mathcal{O}_{\widehat{E}_\infty^\flat}, \qquad R := R^+\left[\frac{1}{\varpi_C^\flat \otimes t}\right].$$

Equivalently,

$$\mathrm{Spa}(R,R^+) = \mathrm{Spa}(R^+,R^+)\setminus V(\varpi_C^\flat \otimes t).$$

In fact, this gives a product interpretation of the tilted Lubin–Tate space: We write

$$\mathrm{Spa}(C^\flat,\mathcal{O}_{C^\flat}) \times_{\mathbb{F}_q} \mathrm{Spa}(\widehat{E}_\infty^\flat,\mathcal{O}_{\widehat{E}_\infty^\flat})$$

for the perfectoid space obtained from the completed tensor product over $\mathbb{F}_q$ as above. Then by an argument similar to the proof of Lemma 4.2(a) in the following section, we have

$$\left(\widetilde{H}_{E,C}^{\mathrm{ad},*}\right)^\flat \cong \mathrm{Spa}(C^\flat,\mathcal{O}_{C^\flat}) \times_{\mathbb{F}_q} \mathrm{Spa}(\widehat{E}_\infty^\flat,\mathcal{O}_{\widehat{E}_\infty^\flat}).$$

When $q = p$, this is precisely the absolute product of perfectoid spaces. For general $q = p^f$, the absolute product is constructed using the completed tensor product over $\mathbb{F}_p$, and the above $\mathbb{F}_q$-linear product is the clopen component determined by the chosen $\mathbb{F}_q$-structures.

3.2. **The space Z_E and Weinstein's theorem.** Following Weinstein, we define

$$Z_E := \widetilde{H}_{E,C}^{\mathrm{ad},*}/E^\times$$

as a sheaf on the pro-étale site of perfectoid spaces over C. Writing

$$E^\times = \mathcal{O}_E^\times \times \pi^{\mathbb{Z}},$$

the action of $\pi^{\mathbb{Z}}$ is totally discontinuous, while

$$\widetilde{H}_{E,C}^{\mathrm{ad},*}/\pi^{\mathbb{Z}} \longrightarrow Z_E$$

is a pro-étale $\mathcal{O}_E^\times$-torsor.

For a group G acting on a space X, write $\mathbf{F\acute{E}t}_G(X)$ for the category of G-equivariant finite étale covers of X.

Theorem 3.1 (Weinstein)**.** *There is a natural equivalence*

$$\mathbf{F\acute{E}t}(Z_E) \simeq \mathbf{F\acute{E}t}(E).$$

In particular,

$$\pi_1^{\text{ét}}(Z_E) \cong G_E.$$

Proof. The result follows from the following chain of equivalences:

$$\begin{aligned}
\mathbf{F\acute{E}t}(Z_E) &\simeq \mathbf{F\acute{E}t}_{E^\times}\left(\widetilde{H}^{\mathrm{ad},*}_{E,C}\right)\\
&\simeq \mathbf{F\acute{E}t}_{E^\times}\left(\left(\widetilde{H}^{\mathrm{ad},*}_{E,C}\right)^\flat\right)\\
&\simeq \mathbf{F\acute{E}t}_{\mathcal{O}_E^\times\times(\varphi^{-1}\circ\varphi_q)^{\mathbb{Z}}}\left(\mathcal{Y}_{C^\flat,E}\widehat{\otimes}_E\widehat{E}_\infty\right)\\
&\simeq \mathbf{F\acute{E}t}_{\mathcal{O}_E^\times\times\varphi^{\mathbb{Z}}}\left(\mathcal{Y}_{C^\flat,E}\widehat{\otimes}_E\widehat{E}_\infty\right)\\
&\simeq \mathbf{F\acute{E}t}_{\mathcal{O}_E^\times}\left(\mathcal{X}_{C^\flat,E}\widehat{\otimes}_E\widehat{E}_\infty\right)\\
&\simeq \mathbf{F\acute{E}t}_{\mathcal{O}_E^\times}(\widehat{E}_\infty)\\
&\simeq \mathbf{F\acute{E}t}(E).
\end{aligned}$$

Since $\widetilde{H}^{\mathrm{ad},*}_{E,C}/\pi^{\mathbb{Z}} \to Z_E$ is a pro-étale $\mathcal{O}_E^\times$-torsor, finite étale covers of Z_E is equivalent to $\mathcal{O}_E^\times$-equivariant finite étale covers of $\widetilde{H}^{\mathrm{ad},*}_{E,C}/\pi^{\mathbb{Z}}$. Since the action of π on $\widetilde{H}^{\mathrm{ad},*}_{E,C}$ is discrete, this is again equivalent to $E^\times$-equivariant finite étale covers of $\widetilde{H}^{\mathrm{ad},*}_{E,C}$, and the first equivalence above holds. The second equivalence is the tilting equivalence, the third follows from the identification of $\left(\widetilde{H}^{\mathrm{ad},*}_{E,C}\right)^\flat$ in the end of the previous subsection. For the fourth equivalence, recall that the absolute q-power Frobenius φ_q is an automorphism of every perfectoid space of characteristic p, and also of every finite étale cover of such a space. Hence, for any $\mathbb{F}_q$-linear automorphism f of a perfectoid space, composing an f-equivariant structure on a finite étale cover with φ_q gives an $(f\circ\varphi_q)$-equivariant structure, and this operation is invertible. Applying this with $f=\varphi^{-1}$ identifies $(\varphi^{-1}\circ\varphi_q)^{\mathbb{Z}}$-equivariant covers with $\varphi^{-1,\mathbb{Z}}$-equivariant covers, or equivalently with $\varphi^{\mathbb{Z}}$-equivariant covers. The fifth is the definition of the adic Fargues–Fontaine curve, while the last two equivalences follow from the geometric simple connectivity of $\mathcal{X}_{C^\flat,E}$ and Galois descent along the Lubin–Tate extension. □

4. Kedlaya's base change lemma

4.1. **Universal geometric simple connectivity.** We first recall the strengthened geometric simple connectivity theorem of Kedlaya [5, Theorem 9.1]. Let F be an algebraically closed nonarchimedean field of characteristic p, let E be a local field with residue field $\mathbb{F}_q$, and let L be a complete algebraically closed overfield of E.

Theorem 4.1 (Kedlaya)**.** *Every finite étale cover of*

$$\mathcal{X}_{F,E}\times_E L$$

splits completely.

Proof. We briefly recall the argument. By a tilting reduction, one may reduce to the case $\mathrm{char}(E)=0$. When L is a completed algebraic closure of E, the assertion follows from the classification of vector bundles on the Fargues–Fontaine curve, as recalled in Section 2.

The general case is obtained by transfinite induction. It is enough to show that, if the assertion holds for L, then it continues to hold after replacing L by the completed algebraic closure L' of the completion of $L(T)$ with respect to an arbitrary multiplicative norm. Such a norm determines a point x of type 2, 3, or 4, in the sense of Berkovich, on the adic affine

line over L. The point x admits a fundamental system of étale neighborhoods D which are smooth one-dimensional affinoid spaces over L.

Let

$$U' \longrightarrow \mathcal{X}_{F,E} \times_E L'$$

be a finite étale cover. This cover spreads out to a finite étale cover

$$U_D \longrightarrow \mathcal{X}_{F,E} \times_E D$$

for a sufficiently small étale neighborhood D of x. The essential input is [5, Lemma 8.4]: after replacing D by a further étale neighborhood $D' \to D$, the cover U_D becomes split. This lemma is obtained from the analysis of the subsidiary radii of the corresponding p-adic differential equations. After base change to L', the original cover therefore splits. The transfinite induction proves the theorem. □

4.2. **Products of perfectoid spaces.** Let **Pfd** denote the category of perfectoid spaces of characteristic p. We recall the construction of absolute products in **Pfd** and the associated partial Frobenius quotients [5, Lemma 9.3].

Lemma 4.2 (Kedlaya)**.** *Let $X_1, \dots, X_n \in \mathbf{Pfd}$.*

(a) *The categorical product*

$$X := X_1 \times \cdots \times X_n$$

exists in **Pfd**. *More precisely, if*

$$X_i = \mathrm{Spa}(A_i, A_i^+)$$

is affinoid perfectoid and $\varpi_i \in A_i$ is a topologically nilpotent unit, put

$$A^+ := \widehat{A_1^+ \otimes_{\mathbb{F}_p} \cdots \otimes_{\mathbb{F}_p} A_n^+},$$

where the completion is taken for the $(\varpi_1, \dots, \varpi_n)$-adic topology, and

$$A := A^+[\varpi_1^{-1}, \dots, \varpi_n^{-1}].$$

Then

$$X_1 \times \cdots \times X_n \cong \left\{ x \in \mathrm{Spa}(A, A^+) \,\middle|\, |\varpi_i(x)| < 1 \text{ for } i = 1, \dots, n \right\}.$$

This description is independent of the choices of the ϖ_i.

(b) *For $i = 1, \dots, n$, let φ_i denote the automorphism of X induced by the absolute p-power Frobenius on X_i. For any fixed i, the quotient*

$$X/\Phi := X/\langle \varphi_1, \dots, \widehat{\varphi_i}, \dots, \varphi_n \rangle$$

exists in **Pfd** *and is canonically independent of the choice of i.*

(c) *If $X_1, \dots, X_n$ are qcqs, then X/Φ is qcqs.*

Proof. For these all assertions, it suffices to treat the case

$$X_i = \mathrm{Spa}(A_i, A_i^+)$$

with each X_i affinoid perfectoid. Choose a topologically nilpotent unit $\varpi_i \in A_i$ for each i. Let

$$A^+ := \widehat{A_1^+ \otimes_{\mathbb{F}_p} \cdots \otimes_{\mathbb{F}_p} A_n^+}$$

where the completion is taken for the $(\varpi_1, \dots, \varpi_n)$-adic topology, and put

$$A := A^+[\varpi_1^{-1}, \dots, \varpi_n^{-1}].$$

Set

$$X := \left\{ x \in \mathrm{Spa}(A, A^+) \,\middle|\, |\varpi_i(x)| < 1 \text{ for } i = 1, \dots, n \right\}.$$

(a) Let Y be a perfectoid space together with morphisms

$$f_i : Y \longrightarrow X_i.$$

The assertion is local on Y, so we may assume

$$Y = \mathrm{Spa}(B, B^+)$$

is affinoid perfectoid. Each f_i corresponds to a morphism of Huber pairs

$$f_i^{\#} : A_i \longrightarrow B$$

sending A_i^+ into B^+. Multiplication in B^+ therefore gives a morphism

$$A_1^+ \otimes_{\mathbb{F}_p} \cdots \otimes_{\mathbb{F}_p} A_n^+ \longrightarrow B^+, \qquad a_1 \otimes \cdots \otimes a_n \longmapsto f_1^{\#}(a_1) \cdots f_n^{\#}(a_n).$$

The elements $f_i^{\#}(\varpi_i)$ are topologically nilpotent units in B. Hence the image of the ideal $(\varpi_1, \dots, \varpi_n)$ is topologically nilpotent, so the preceding morphism is continuous for the $(\varpi_1, \dots, \varpi_n)$-adic topology and extends uniquely to

$$A^+ \longrightarrow B^+.$$

Since each $f_i^{\#}(\varpi_i)$ is invertible in B, this further extends uniquely to

$$A \longrightarrow B.$$

We obtain a morphism

$$f : Y \longrightarrow \mathrm{Spa}(A, A^+).$$

For every $y \in Y$,

$$|\varpi_i(f(y))| = |f_i^{\#}(\varpi_i)(y)| < 1,$$

so the image of f lies in X. By construction, the composite $Y \to X \to X_i$ is f_i for every i. Conversely, any morphism $Y \to X$ gives the morphisms to the X_i by composition. These constructions are inverse to each other, giving

$$\mathrm{Hom}_{\mathbf{Pfd}}(Y, X) \cong \prod_{i=1}^{n} \mathrm{Hom}_{\mathbf{Pfd}}(Y, X_i).$$

Thus X is the categorical product of the X_i in **Pfd**.

(b) Fix $i = 1$ for the notation. For $x \in X$ and $j = 2, \dots, n$, put

$$r_j(x) := \frac{\log |\varpi_j(x)|}{\log |\varpi_1(x)|} \in \mathbb{R}_{>0}.$$

Choose rational numbers $a_j, b_j > 0$ such that

$$a_j \le r_j(x) \le b_j < p a_j.$$

Then x belongs to the rational neighborhood

$$U = \left\{ y \in X \,\middle|\, |\varpi_1(y)|^{b_j} \le |\varpi_j(y)| \le |\varpi_1(y)|^{a_j} \text{ for } j = 2, \dots, n \right\}.$$

Under φ_j, the quantity r_j is multiplied by p. Hence the translates of U by

$$\Phi = \langle \varphi_2, \dots, \varphi_n \rangle$$

are pairwise disjoint, because

$$b_j < p a_j.$$

Thus the action of Φ on X is totally discontinuous. The quotients of these neighborhoods by their stabilizers glue to a perfectoid space X/Φ, proving (b).

(c) If the X_i are qcqs, one may choose finitely many such neighborhoods whose Φ-translates cover X. Their pairwise intersections are again of the same form. It follows that X/Φ is quasicompact and separated, hence qcqs.

□

In particular, Proposition 2.4 give the following identification:

Remark. Let $E = \mathbb{F}_q((\pi))$, let F be an algebraically closed nonarchimedean field of characteristic p containing $\mathbb{F}_q$, and let L be a complete algebraically closed overfield of E. Then

$$\mathcal{X}_{F,E} \times_E L \cong (\mathrm{Spa}(F, \mathcal{O}_F) \times \mathrm{Spa}(L, \mathcal{O}_L))/\Phi,$$

where the product is the categorical product in **Pfd** and Φ is the corresponding partial Frobenius action. Consequently, the universal geometric simple connectivity theorem implies that

$$\pi_1^{\mathrm{prof}}((\mathrm{Spa}(F, \mathcal{O}_F) \times \mathrm{Spa}(L, \mathcal{O}_L))/\Phi) = 1.$$

4.3. **The base change lemma.** Fix an algebraically closed nonarchimedean field F of characteristic p. For $X \in \mathbf{Pfd}$, put

$$X_F := X \times \mathrm{Spa}(F, \mathcal{O}_F),$$

where the product is taken in **Pfd**, and let

$$\varphi_F : X_F \longrightarrow X_F$$

be the automorphism induced by the absolute p-power Frobenius on the second factor.

Lemma 4.3 (Kedlaya's base change lemma)**.** *The base extension functor*

$$\mathbf{F\acute{E}t}(X) \longrightarrow \mathbf{F\acute{E}t}(X_F/\varphi_F)$$

is an equivalence of categories.

Proof. This is [5, Lemma 9.5]. Both

$$X \longmapsto \mathbf{F\acute{E}t}(X) \qquad \text{and} \qquad X \longmapsto \mathbf{F\acute{E}t}(X_F/\varphi_F)$$

are stacks for the étale topology. It therefore suffices to verify the assertion on geometric points.

Let

$$X = \mathrm{Spa}(L, L^+)$$

be a geometric point, with L algebraically closed of characteristic p. Choose a topologically nilpotent unit $t \in L$ and put

$$E := \mathbb{F}_p((t)).$$

Then L is a complete algebraically closed overfield of E. By the preceding remark,

$$X_F/\varphi_F \cong \mathcal{X}_{F,E} \times_E L.$$

The universal geometric simple connectivity theorem therefore implies that every finite étale cover of X_F/φ_F splits completely. Since L is algebraically closed, every finite étale cover of X also splits completely. Thus both categories are canonically equivalent to the category of finite sets, and

$$\mathbf{F\acute{E}t}(X) \xrightarrow{\sim} \mathbf{F\acute{E}t}(X_F/\varphi_F).$$

Étale descent then gives the equivalence for arbitrary $X \in \mathbf{Pfd}$. □

5. Completion and infinite Galois descent

Let K be a complete nonarchimedean field of rank one, let K_∞/K be a Galois algebraic extension, and put

$$\Gamma := \mathrm{Gal}(K_\infty/K), \qquad L := \widehat{K_\infty}.$$

Since K is complete, its valuation is henselian and hence extends uniquely to K_∞. In particular, every element of Γ acts isometrically on K_∞ and therefore extends uniquely to a continuous automorphism of L.

If a topological field M is equipped with a continuous action of Γ, we write $\mathbf{F\acute{E}t}_\Gamma(M)$ for the category of finite étale M-algebras equipped with a compatible continuous semilinear Γ-action. Equivalently, we may regard this as the category of Γ-equivariant finite étale covers of $\mathrm{Spa}(M, \mathcal{O}_M)$.

5.1. **Completion.** We first record that passing from K_∞ to its completion does not change its finite étale theory.

Lemma 5.1. *The completion functor induces an equivalence*

$$\mathbf{F\acute{E}t}(K_\infty) \xrightarrow{\sim} \mathbf{F\acute{E}t}(L).$$

Moreover, this equivalence is Γ-equivariant and hence induces an equivalence

$$\mathbf{F\acute{E}t}_\Gamma(K_\infty) \xrightarrow{\sim} \mathbf{F\acute{E}t}_\Gamma(L).$$

Proof. The field K_∞ is henselian, since it is an algebraic extension of the henselian valued field K. In the terminology of Berkovich, it is therefore quasicomplete. By [8, Propositions 2.4.1 and 2.4.3], completion induces an equivalence between the categories of finite separable extensions of K_∞ and of L.

More explicitly, if M/K_∞ is finite separable, then its valuation extends uniquely from K_∞, and

$$\widehat{M} \cong M \otimes_{K_\infty} L, \qquad [\widehat{M} : L] = [M : K_\infty].$$

Conversely, every finite separable extension of L is obtained in this way from a finite separable extension of K_∞; the essential surjectivity follows from Krasner's lemma. Taking finite products gives the asserted equivalence for finite étale algebras.

The construction is functorial with respect to isometric automorphisms of K_∞. Thus for $\gamma \in \Gamma$, completion commutes with the pullback by γ, and the equivalence above is Γ-equivariant. A Γ-equivariant structure and its cocycle condition are therefore transported uniquely across the equivalence, giving

$$\mathbf{F\acute{E}t}_\Gamma(K_\infty) \xrightarrow{\sim} \mathbf{F\acute{E}t}_\Gamma(L).$$

□

5.2. **Infinite Galois descent.** We next recall the Galois descent statement that will be used in the proof of the main theorem.

Lemma 5.2. *Base extension induces an equivalence*

$$\mathbf{F\acute{E}t}(K) \xrightarrow{\sim} \mathbf{F\acute{E}t}_\Gamma(K_\infty).$$

Proof. The morphism

$$\mathrm{Spec}(K_\infty) \longrightarrow \mathrm{Spec}(K)$$

is faithfully flat and quasi-compact. Finite étale morphisms satisfy effective fpqc descent; see [10, Exposé IX, §§4–5]. Hence finite étale K-algebras are equivalent to finite étale K_∞-algebras equipped with descent data relative to K_∞/K.

Since K_∞/K is Galois, such descent data may equivalently be expressed as continuous semilinear Γ-actions. We briefly recall why the fact that Γ may be infinite causes no difficulty. Write

$$K_\infty = \bigcup_M M,$$

where M/K runs through the finite Galois subextensions. A finite étale K_∞-algebra is of finite presentation, so it descends to some finite stage M. Likewise, all morphisms occurring in a descent datum descend after enlarging M. At a finite Galois stage, the usual identification

$$M \otimes_K M \cong \prod_{\sigma \in \mathrm{Gal}(M/K)} M$$

identifies a descent datum with a semilinear $\mathrm{Gal}(M/K)$-action satisfying the cocycle condition. Passing to the filtered union identifies descent data over K_∞/K with continuous semilinear Γ-actions. Thus

$$\mathbf{F\acute{E}t}(K) \xrightarrow{\sim} \mathbf{F\acute{E}t}_\Gamma(K_\infty).$$

□

6. Proof of the main theorem

6.1. **The construction and the main theorem.** Let E be a complete rank-one nonarchimedean field of mixed characteristic $(0,p)$. Fix an algebraic closure $\overline{E}$ and put

$$C := \widehat{\overline{E}}.$$

Let E_∞/E be a Galois algebraic extension contained in $\overline{E}$, with profinite Galois group

$$\Gamma := \mathrm{Gal}(E_\infty/E).$$

Suppose that

$$L := \widehat{E_\infty}$$

is perfectoid, and put

$$F_\infty := L^\flat.$$

The action of Γ on E_∞ extends uniquely to a continuous faithful action on L, and hence induces a continuous faithful action on F_∞.

Consider the categorical product in **Pfd**

$$P^\flat := \mathrm{Spa}(C^\flat, \mathcal{O}_{C^\flat}) \times \mathrm{Spa}(F_\infty, \mathcal{O}_{F_\infty}).$$

Let φ_C and φ_{F_∞} denote the partial absolute Frobenii on the first and second factors, respectively, and put

$$V^\flat_{F_\infty} := P^\flat/\varphi^{\mathbb{Z}}_{F_\infty}, \qquad V^\flat_C := P^\flat/\varphi^{\mathbb{Z}}_C.$$

By Lemma 4.2(b), there is a canonical isomorphism

$$\iota : V^\flat_{F_\infty} \xrightarrow{\sim} V^\flat_C,$$

functorial in both factors. The group Γ acts on the second factor of $P^\flat$ and commutes with both partial Frobenii; in particular, the preceding isomorphism is Γ-equivariant.

The structural morphism

$$V_{F_\infty}^\flat \longrightarrow \mathrm{Spa}(C^\flat, \mathcal{O}_{C^\flat})$$

allows us to untilt $V_{F_\infty}^\flat$ over C. Define

$$V := \left(V_{F_\infty}^\flat\right)_C^\sharp.$$

The action of Γ on $V_{F_\infty}^\flat$ untilts to a continuous C-linear action on V. Finally, define

$$Z := Z(E_\infty/E; C) := V/\Gamma$$

as the sheaf quotient on the pro-étale site of perfectoid spaces over C.

Theorem 6.1. *There is a natural equivalence of Galois categories*

$$\mathbf{F\acute{E}t}(Z) \xrightarrow{\sim} \mathbf{F\acute{E}t}(E).$$

In particular, after choosing compatible geometric base points,

$$\pi_1^{\text{ét}}(Z) \cong G_E.$$

Proof. The desired equivalence is the composite

$$\mathbf{F\acute{E}t}(Z) \overset{(1)}{\simeq} \mathbf{F\acute{E}t}_\Gamma(V) \overset{(2)}{\simeq} \mathbf{F\acute{E}t}_\Gamma(V_{F_\infty}^\flat) \overset{(3)}{\simeq} \mathbf{F\acute{E}t}_\Gamma(F_\infty)$$
$$\overset{(4)}{\simeq} \mathbf{F\acute{E}t}_\Gamma(L) \overset{(5)}{\simeq} \mathbf{F\acute{E}t}_\Gamma(E_\infty) \overset{(6)}{\simeq} \mathbf{F\acute{E}t}(E).$$

For (1), we first show that the action of Γ on V is free on geometric points. Let C'/C be a complete algebraically closed extension and suppose that $\gamma \in \Gamma$ fixes a point

$$\overline{v} \in V(C').$$

By the tilting equivalence and the canonical isomorphism ι, we may regard $\overline{v}$ as a point of $V_C^\flat(C'^\flat)$. After replacing C' by a further complete algebraically closed extension if necessary, choose a lift

$$\overline{p} \in P^\flat(C'^\flat)$$

of $\overline{v}$. Since

$$V_C^\flat = P^\flat/\varphi_C^{\mathbb{Z}},$$

the equality $\gamma\overline{v} = \overline{v}$ implies

$$\gamma\overline{p} = \varphi_C^m \overline{p}$$

for some $m \in \mathbb{Z}$. Projecting to the second factor gives a continuous embedding

$$j : F_\infty \hookrightarrow C'^\flat$$

such that

$$j \circ \gamma = j,$$

because φ_C acts only on the first factor. Since j is injective, γ acts trivially on F_∞. The action of Γ on $F_\infty = L^\flat$ is faithful, hence $\gamma = 1$. Thus the action of Γ on $V(C')$ is free for every complete algebraically closed extension C'/C.

By [6, Proposition 4.3.2], the quotient map

$$V \longrightarrow Z = V/\Gamma$$

is therefore a pro-étale Γ-torsor. Effective descent for finite étale covers along such torsors [6, Proposition 4.3.3] gives

$$\mathbf{F\acute{E}t}(Z) \xrightarrow{\sim} \mathbf{F\acute{E}t}_{\Gamma}(V).$$

The equivalence (2) is the tilting equivalence

$$\mathbf{F\acute{E}t}(V) \simeq \mathbf{F\acute{E}t}(V^{\flat}_{F_\infty}).$$

Its functoriality with respect to automorphisms identifies the corresponding Γ-equivariant structures.

For (3), use first the canonical Γ-equivariant isomorphism

$$V^{\flat}_{F_\infty} \xrightarrow{\sim} V^{\flat}_{C}.$$

Apply Lemma 4.3, with

$$X = \mathrm{Spa}(F_\infty, \mathcal{O}_{F_\infty})$$

and with the algebraically closed perfectoid field $C^{\flat}$. The product occurring there is precisely $P^{\flat}$, and quotienting by the partial Frobenius on the $C^{\flat}$-factor gives $V^{\flat}_{C}$. Hence

$$\mathbf{F\acute{E}t}(F_\infty) \xrightarrow{\sim} \mathbf{F\acute{E}t}(V^{\flat}_{C}) \xrightarrow{\sim} \mathbf{F\acute{E}t}(V^{\flat}_{F_\infty}).$$

These equivalences are functorial in automorphisms of F_∞, and therefore induce

$$\mathbf{F\acute{E}t}_{\Gamma}(V^{\flat}_{F_\infty}) \xrightarrow{\sim} \mathbf{F\acute{E}t}_{\Gamma}(F_\infty).$$

The equivalence (4) follows from the tilting equivalence for the perfectoid fields

$$F_\infty = L^{\flat}.$$

Again, functoriality identifies the Γ-actions.

Finally, the equivalence (5) follows from Lemma 5.1 and (6) follows from Lemma 5.2. □

6.2. **The case of $\mathbb{Q}_p\{\{X\}\}$.** We now apply Theorem 6.1 to

$$E = \mathbb{Q}_p\{\{X\}\}.$$

Choose compatible systems of p-power roots of unity and p-power roots of X, and put

$$E_n := E(\mu_{p^n}, X^{1/p^n}), \qquad E_\infty := \bigcup_{n\geq 1} E_n, \qquad \Gamma_E := \mathrm{Gal}(E_\infty/E).$$

This is the one-dimensional p-basis case of the Kummer tower considered by Scholl [11, §2.3]. It is strictly deeply ramified; consequently, its completion

$$L := \widehat{E_\infty}$$

is a perfectoid field by [7, Proposition 6.6.6]; see also [9, Remark 3.3].

Put

$$F_\infty := L^{\flat}, \qquad C := \widehat{\overline{E}},$$

and form

$$P^{\flat} = \mathrm{Spa}(C^{\flat}, \mathcal{O}_{C^{\flat}}) \times \mathrm{Spa}(F_\infty, \mathcal{O}_{F_\infty}), \qquad V^{\flat}_{F_\infty} = P^{\flat}/\varphi^{\mathbb{Z}}_{F_\infty}.$$

Let

$$V := \big(V^{\flat}_{F_\infty}\big)^{\sharp}_{C}, \qquad Z_E := V/\Gamma_E.$$

Then Theorem 6.1 gives

$$\mathbf{F\acute{E}t}(Z_E) \simeq \mathbf{F\acute{E}t}\big(\mathbb{Q}_p\{\{X\}\}\big),$$

and hence

$$\pi_1^{\text{ét}}(Z_E) \cong G_{\mathbb{Q}_p\{\{X\}\}}.$$

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